

CHIVENOR, United Kingdom

WMO: 037070

Lat: 51.0892N

Lon: 4.1486W

Elev: 8

StdP: 101.23

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-1.8	-0.4	-5.3	2.4	0.5	-3.8	2.7	1.6	17.3	8.1	15.3	8.9	3.6	90	0.653

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.7	24.7	18.0	22.6	17.1	21.0	16.5	19.0	22.9	18.1	21.1	17.4	19.9	5.2	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.6	12.6	20.2	16.9	12.1	19.3	16.2	11.5	18.7	53.6	23.0	51.0	21.1	48.8	20.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.1	12.6	11.3	DB	-4.3	28.0	1.7	1.8	-5.5	29.3	-6.5	30.4	-7.5	31.4	-8.7	32.8	(4)
(5)			WB	-4.9	20.9	1.7	1.3	-6.0	21.9	-7.0	22.6	-7.9	23.4	-9.1	24.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.3	6.5	6.6	7.8	9.5	12.3	15.0	16.8	16.7	15.0	12.4	9.2	7.2
	DBStd	4.54	3.00	2.73	2.55	2.49	2.40	2.20	1.97	1.90	1.95	2.53	2.76	3.28
	HDD10.0	478	112	98	76	36	6	0	0	0	0	8	46	95
	HDD18.3	2596	368	329	327	263	187	104	57	57	102	183	274	345
	CDD10.0	945	3	2	6	23	78	149	211	209	150	84	22	8
	CDD18.3	22	0	0	0	0	0	3	9	8	2	0	0	0
	CDH23.3	132	0	0	0	1	4	32	58	33	4	1	0	0
	CDH26.7	20	0	0	0	0	0	6	9	6	0	0	0	0
Wind	WSAvg	5.3	6.1	5.9	5.6	5.1	5.0	4.7	4.4	4.5	4.8	5.4	5.5	6.1
Precipitation	PrecAvg	1035	111	84	65	59	62	70	73	81	78	116	122	115
	PrecMax	1376	194	185	114	140	126	167	218	148	147	212	212	227
	PrecMin	796	18	20	20	16	8	15	11	14	17	39	60	41
	PrecStd	142	42	40	25	32	28	41	43	39	30	38	40	50
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.2	13.9	16.8	21.2	23.7	27.0	27.8	27.1	23.8	19.7	15.7	14.4
		MCWB	11.7	11.5	11.6	14.3	16.5	18.8	19.0	19.2	18.5	16.2	14.0	12.8
	2%	DB	12.3	12.1	14.1	17.5	20.8	23.2	24.9	23.4	21.2	17.7	14.7	13.3
		MCWB	11.1	10.4	10.6	12.1	15.4	17.1	17.7	18.1	16.8	15.1	13.2	12.1
	5%	DB	11.5	11.2	12.5	15.1	18.4	20.8	22.4	21.4	19.7	16.9	14.0	12.5
		MCWB	10.4	9.7	10.1	11.0	13.8	15.7	17.1	17.2	16.0	14.6	12.6	11.4
	10%	DB	10.6	10.5	11.4	13.5	16.5	19.0	20.6	20.1	18.5	16.0	13.2	11.6
		MCWB	9.5	9.2	9.5	10.4	13.0	15.1	16.6	16.5	15.4	14.2	11.8	10.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.2	12.0	12.4	14.6	17.6	19.7	20.1	20.3	19.2	17.1	14.6	13.1
		MCDB	12.7	13.2	15.0	19.7	22.0	25.6	25.8	24.6	22.2	19.1	15.3	13.8
	2%	WB	11.4	10.8	11.4	12.8	15.8	17.6	18.7	18.9	17.8	15.9	13.6	12.4
		MCDB	12.1	11.7	13.3	16.2	19.6	21.9	23.1	22.1	20.0	17.1	14.4	13.2
	5%	WB	10.5	10.1	10.6	11.8	14.6	16.4	17.8	18.0	16.9	15.2	12.9	11.6
		MCDB	11.3	11.0	12.0	14.2	17.6	19.7	21.1	20.3	18.8	16.4	13.7	12.4
	10%	WB	9.7	9.4	9.9	10.9	13.6	15.6	17.1	17.3	16.1	14.5	12.0	10.7
		MCDB	10.5	10.4	11.2	13.1	15.9	18.2	19.7	19.4	18.0	15.8	13.1	11.6
Mean Daily Temperature Range	5% DB	MDBR	4.8	5.2	6.1	7.2	7.3	6.9	6.7	6.5	6.8	5.9	5.2	4.9
		MCDBR	4.8	5.5	7.9	10.1	10.6	10.7	10.8	9.0	8.7	6.2	4.9	4.5
		MCWBR	4.7	4.5	5.4	6.0	6.2	5.9	5.4	4.8	5.3	4.4	4.2	4.5
	5% WB	MCDBR	4.8	4.9	6.5	8.5	9.3	8.9	8.5	7.5	7.1	5.4	4.5	4.3
		MCWBR	5.0	4.5	5.0	5.5	5.8	5.3	5.0	4.6	5.1	4.4	4.4	4.6
Clear-Sky Solar Irradiance	taub	0.334	0.347	0.376	0.390	0.397	0.386	0.387	0.373	0.366	0.356	0.332	0.324	
	taud	2.341	2.370	2.302	2.290	2.309	2.373	2.374	2.431	2.439	2.434	2.393	2.314	
	Ebn at Noon	672	775	816	849	858	870	862	858	822	755	672	616	
	Edh at Noon	66	82	105	119	122	116	114	102	91	77	62	58	
All-Sky Solar Radiation	RadAvg	0.78	1.54	2.73	4.34	5.29	5.77	5.33	4.46	3.34	1.87	0.97	0.61	
	RadStd	0.07	0.15	0.29	0.35	0.32	0.53	0.42	0.32	0.24	0.14	0.08	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page